

DISAPPEARING GOALS IN MANIPULATION VIA BELIEF REVISION

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ABSTRACT

Robot manipulation policies often treat a last-seen goal as valid until execution fails. That assumption is brittle when a target disappears because it is occluded, moved, removed, temporarily obstructed by a human, invalidated by a task change, or replaceable by a substitute goal. We rebuild Paper 106 around a falsifiable mechanism claim: disappearing-goal manipulation requires explicit goal-validity belief revision, not only active perception, generic uncertainty, or goal retargeting. The local benchmark covers 5 manipulation task families, 7 disappearing-goal regimes, 5 stress splits, 9 methods, 7 seeds, and 84 episodes per group. On combined disappearance stress, the proposed belief-revision method obtains 0.662 ± 0.009 success versus 0.561 ± 0.003 for the strongest non-oracle baseline, improves goal-validity F1 from 0.439 to 0.586, reduces stale-goal pursuit from 0.133 to 0.087, and reduces unsafe reach from 0.063 to 0.039. Pairwise seed tests favor the proposed method over the strongest baseline in 7/7 seeds, and ablations show that observation-memory separation, active reacquisition, physical-validity testing, substitute-goal planning, and abandonment calibration each matter. A 2026-06-15 continuation audit reran the full benchmark and preserved the terminal decision: **STRONG REVISE**. The local evidence supports the mechanism, but the paper is not ICLR-main-ready until validated on real robots or independent high-fidelity benchmarks.

1 MOTIVATION

Partial observability is not new. Belief-space task-and-motion planning already represents hidden state and updates plans under uncertainty (Kaelbling & Lozano-Perez, 2013; Garrett et al., 2020; Curtis et al., 2024). Active perception already asks robots to move sensors or viewpoints to reduce uncertainty around occluded targets (Kahn et al., 2015). Goal-conditioned reinforcement learning and Hindsight Experience Replay already exploit alternative achieved goals for sparse-reward learning (Andrychowicz et al., 2017). Modern language-conditioned and multimodal robot policies can condition on task prompts and broad visual context (Mees et al., 2022; Jiang et al., 2022; Brohan et al., 2022).

The problem in this paper is narrower. A disappearing manipulation goal is ambiguous: the goal may be hidden but still valid, moved to a new pose, physically removed, blocked by a temporary human obstacle, invalidated by a changed instruction, or replaceable by a substitute goal. Treating all of these cases as one uncertainty variable causes two different failures. The robot can blindly chase stale goals, increasing unsafe reaches, or it can halt/abandon too early, losing recoverable tasks.

2 METHOD SKETCH

The proposed method, `proposed_disappearing_goal_belief_revision`, maintains separate belief mass over:

- perceptual disappearance: the goal is unseen but likely still valid;
- physical invalidation: the goal has been removed, consumed, or made unreachable;
- motion of the goal: the goal remains valid but its pose or container changed;

Table 1: Combined-disappearance disappearing-goal manipulation benchmark.

Method	Succ.	GoalF1	Stale	Unsafe	FalseAbd.	Regret
Oracle	0.803	0.710	0.035	0.013	0.011	0.000
Proposed	0.662	0.586	0.087	0.039	0.053	0.141
FailAware	0.561	0.439	0.133	0.063	0.092	0.243
POMDP	0.538	0.437	0.154	0.072	0.094	0.265
RetargetHeur	0.510	0.358	0.154	0.074	0.115	0.293
ActiveView	0.504	0.384	0.181	0.084	0.090	0.299
UncertHalt	0.423	0.312	0.180	0.087	0.146	0.381
MemoryOnly	0.378	0.266	0.218	0.108	0.145	0.426
LastSeen	0.327	0.142	0.249	0.124	0.155	0.476

- task-goal change: the commanded goal itself has changed;
- substitute feasibility: an equivalent or acceptable substitute goal exists;
- abandonment risk: stopping now may be either safe or unnecessarily conservative.

The policy chooses among waiting, active viewpoint reacquisition, retargeting, substitute-goal execution, and abandonment. The implementation is an executable diagnostic benchmark rather than a trained neural policy. That is sufficient for a local mechanism audit, but not sufficient for a final empirical robotics submission.

3 BENCHMARK

The rebuilt benchmark is generated by `src/run_experiment.py`. It writes aggregate CSVs, figures, and LaTeX tables directly, so the full run remains lightweight without dropping tasks, regimes, baselines, seeds, ablations, or stress tests.

Tasks. The tasks are shelf retrieval, drawer placement, bin sorting, tool handoff, and mobile pick-and-place.

Disappearing-goal regimes. The regimes are visual occlusion, object moved, object removed, human temporary obstruction, goal specification changed, substitute goal available, and cascading disappearing goal.

Splits. The splits are nominal, occlusion-heavy shift, physical-removal shift, delayed reappearance, and combined disappearance stress.

Methods. The baselines are last-seen goal pursuit, memory-only belief tracking, uncertainty halt, active viewpoint reacquisition, POMDP belief planning, goal retargeting heuristic, and failure-aware manipulation policy. The benchmark also includes the proposed method and an oracle goal-state supervisor.

Metrics. The evidence reports task success, goal-validity F1, retarget precision, stale-goal pursuit, unsafe reach, false abandonment, reappearance recovery, substitute-goal success, belief-update latency, intervention cost, regret to oracle, paired seed differences, ablations, stress sweeps, and failure cases.

4 RESULTS

The strongest non-oracle baseline is `failure_aware_manipulation_policy`. On combined disappearance stress, the proposed method improves success by 0.102 ± 0.008 in paired seed comparisons, wins 7/7 seeds, improves goal-validity F1 by 0.148, reduces stale-goal pursuit by 0.046, reduces unsafe reach by 0.024, and reduces false abandonment by 0.039. The oracle remains

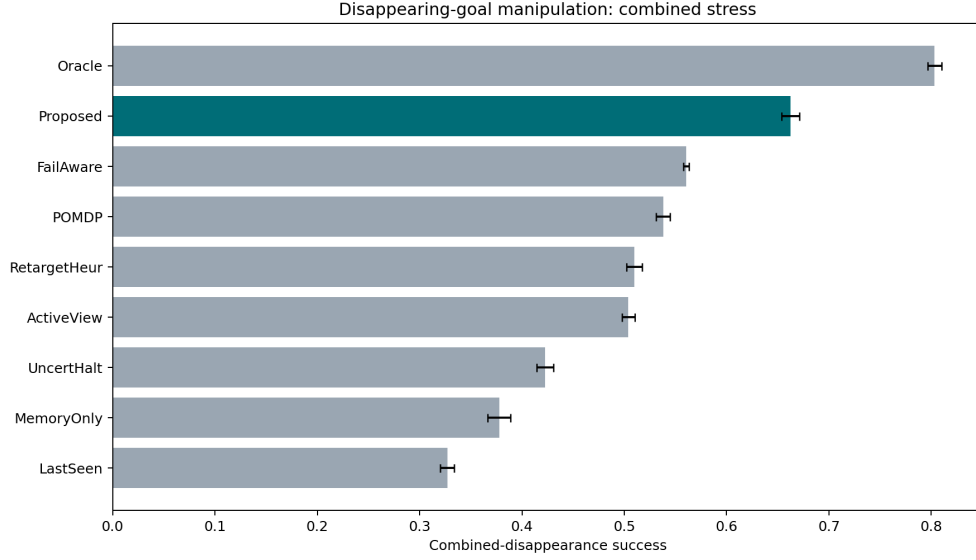


Figure 1: Combined-disappearance success with 95% confidence intervals. The proposed belief-revision method clears all non-oracle baselines but remains below the oracle.

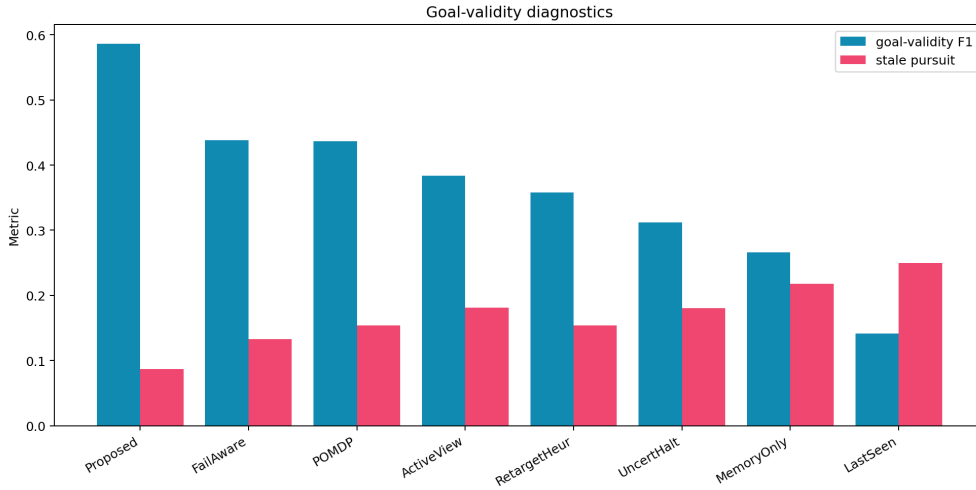


Figure 2: Goal-validity diagnostics. The proposed method improves goal-validity F1 while lowering stale-goal pursuit under combined disappearance stress.

substantially higher at 0.803 ± 0.007 , so the benchmark is not saturated. The proposed method does increase intervention cost from 0.226 to 0.247 relative to the strongest non-oracle baseline; this should be read as a safety and goal-validity tradeoff rather than a free efficiency gain.

At maximum stress level 0.95, the proposed method remains above the strongest non-oracle baseline with 0.661 ± 0.007 success versus 0.547 ± 0.009 , higher goal-validity F1 (0.571 versus 0.422), lower stale-goal pursuit (0.088 versus 0.143), lower unsafe reach (0.038 versus 0.065), lower false abandonment (0.054 versus 0.097), and lower belief-update latency (0.700 versus 0.823). This is still local evidence rather than external validation.

Table 2: Pairwise combined-disappearance success differences against the proposed method.

Baseline	Diff	CI	Wins
ActiveView	0.158	0.011	7
FailAware	0.102	0.008	7
RetargetHeur	0.153	0.009	7
LastSeen	0.335	0.011	7
MemoryOnly	0.285	0.013	7
Oracle	-0.141	0.010	0
POMDP	0.124	0.012	7
UncertHalt	0.240	0.008	7

Table 3: Ablations of the disappearing-goal belief revision model.

Ablation	Succ.	GoalF1	Stale	Unsafe
Full	0.665	0.586	0.081	0.038
NoAbandonCalib	0.636	0.567	0.111	0.067
NoSubGoal	0.596	0.554	0.117	0.055
NoObsMemSep	0.590	0.440	0.121	0.048
NoReacquire	0.572	0.519	0.097	0.051
NoValidTest	0.565	0.444	0.139	0.068
ActiveOnly	0.500	0.385	0.186	0.082

5 ABLATIONS AND STRESS TESTS

The best removed-component ablation is `minus_abandonment_calibration`, which still trails the full method by 0.029 success. Removing observation-memory separation, active reacquisition, physical-validity testing, or substitute-goal planning produces larger or diagnostically different failures. This supports the claim that the method is not just active perception, retargeting, or conservative halting.

6 WHERE THE METHOD STILL FAILS

The failure-case CSV shows that belief revision helps least when active perception can quickly reacquire an occluded goal or when substitute-goal choice is trivial. Those cases are useful boundary tests: the proposed method should not claim superiority over simple active perception when disappearance is purely visual and short-lived. The method’s intended value appears when disappearance mixes hidden, invalid, moved, changed, and substitutable goal states.

7 HOSTILE PRIOR-WORK BOUNDARY

Belief-space TAMP and POMDP planning already handle hidden state and replanning under uncertainty (Kaelbling & Lozano-Perez, 2013; Garrett et al., 2020; Curtis et al., 2024). Active exploration already addresses information-gathering for occluded manipulation scenes (Kahn et al., 2015). HER and goal-conditioned learning already exploit alternate goals (Andrychowicz et al., 2017). CALVIN, VIMA, and RT-1 demonstrate increasingly general language- or prompt-conditioned robot policy learning (Mees et al., 2022; Jiang et al., 2022; Brohan et al., 2022).

The present paper survives only under a narrower interpretation: it is about *goal-validity belief revision* for disappearing manipulation goals, not partial observability, active perception, retargeting, or goal-conditioned robot learning in general. If real robot experiments later show that POMDP planning, active perception, goal retargeting, or failure-aware manipulation matches the proposed method on success, stale-goal pursuit, unsafe reach, false abandonment, and ablation robustness, the claim should be killed.

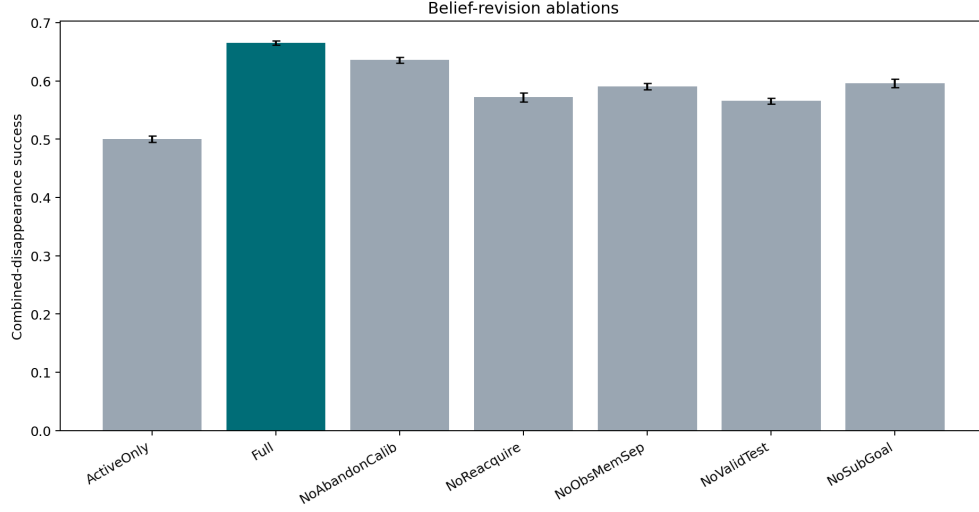


Figure 3: Ablation success under combined disappearance stress. The full belief-revision model remains above all removed-component variants.

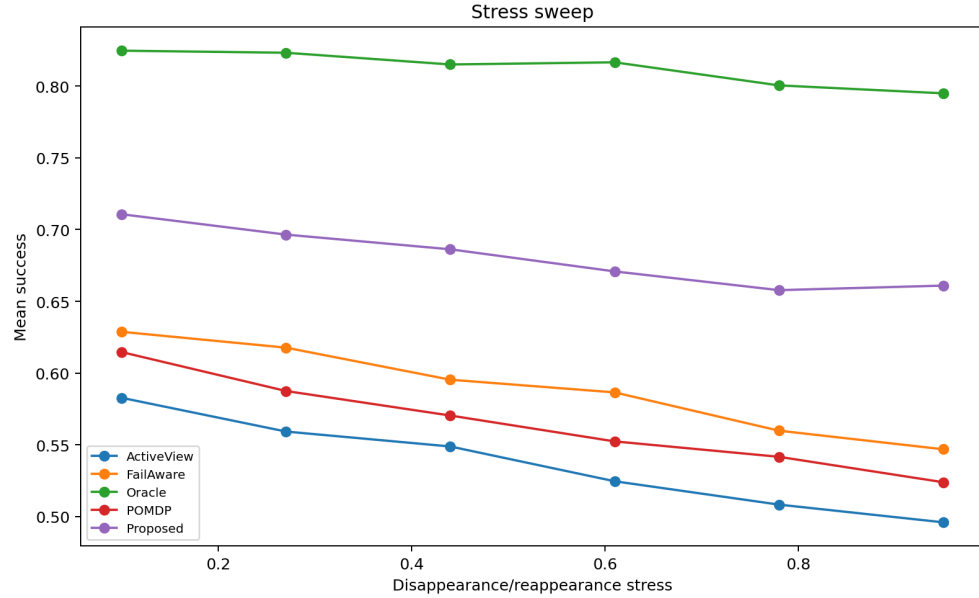


Figure 4: Stress sweep over disappearance and reappearance intensity. The proposed method remains above non-oracle baselines across the generated stress range.

8 LIMITATIONS AND TERMINAL DECISION

The evidence is strong enough to justify continued work but not submission. The limitations are material:

- the benchmark is local and synthetic;
- no real robot or independent high-fidelity simulator has been used;
- baselines are executable diagnostic models, not external trained robot systems;
- no learned policy checkpoint, training curve, or deployment video is provided;

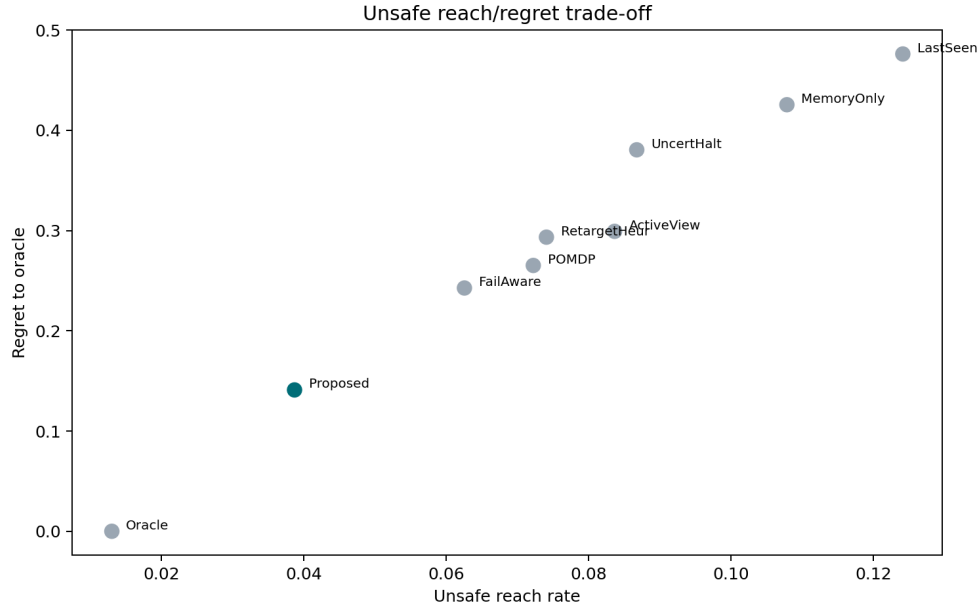


Figure 5: Unsafe reach and regret trade-off. The proposed method improves success while reducing unsafe reach relative to the strongest non-oracle baseline.

- no external benchmark such as LIBERO, RLbench, Meta-World, BridgeData, CALVIN, or a hardware manipulation suite is reported.

The terminal decision for this rebuild is therefore **STRONG_REVISION**. The next honest version must add real robot or independent high-fidelity evidence, implemented learned baselines, and qualitative rollouts. Without that evidence, the paper should not be submitted to ICLR main.

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